

Program : Diploma in Communication and Computer Networking	
Course Code : 6321C	Course Title: IoT Fundamentals and Design
Semester : 6	Credits: 4
Course Category: Engineering Science	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- To assess the vision of IoT.
- To classify Real World IoT applications in various Domains.
- To understand design methodology for IoT platforms.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic knowledge computer networks and related courses.	4322	Computer Networks	4

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Interpret the vision of IoT from a global context.	15	Understand
CO2	Illustrate the application of IoT in various Domains.	15	Applying
CO3	Interpretation of different IoT platforms design methodology.	14	Applying
CO4	Illustration of IoT Physical Devices.	14	Applying
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2	1				
CO2	3	1	2				
CO3	3	2	1				
CO4	3	2	1				

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Interpret the vision of IoT from a global context.		
M1.01	To learn the basics of IoT	4	Understanding
M1.02	Explain the IoT Protocols and Models	4	Understanding
M1.03	Learn about IoT enabling Technologies	3	Applying
M1.04	To learn the different IoT Levels and Deployment templates	4	Applying
Contents: Definition and characteristics of IoT, Physical design of IoT, Things in IoT, IoT Protocols, Logical Design of IoT, IoT functional blocks, IoT communication Models, IoT communication API's, IoT enabling Technologies-Wireless sensor networks, IoT Levels and Deployment templates – IoT Level-1, IoT Level-2, IoT Level-3, IoT Level-4, IoT Level-5, IoT Level-6			
CO2	Illustrate the application of IoT in various Domains.		
M2.01	To understand the Domain specific IoT	7	Understanding
M2.02	To understand the examples of home automation	8	Applying
	Series Test –I	1	

Contents: Domain specific IoT Introduction, Home automation- Smart lighting, smart appliances, intrusion detection, smoke for gas detectors; Cities- Smart Parking, Smart lighting, Smart Roads, Structural Health Monitoring, surveillance, Emergency Response; Environment- Weather monitoring, air pollution monitoring, noise pollution monitoring, forest fire detection, river flood's detection.			
CO3	Interpretation of different IoT platforms design methodology.		
M3.01	To learn the IoT Design and Methodology	4	Understanding
M3.02	To learn its Purpose and requirements specification	4	Understanding
M3.03	To learn about Device and component integration	3	Applying
M3.04	To understand application development.	3	Applying
Contents: IoT Platforms Design Methodology Introduction, IoT Design and Methodology- Purpose and requirements specification, Process specification, Domain model specification, Information model specification, service specification, IoT level specification, functional view specification, Operational view specification, Device and component integration, application development.			
CO4	Illustration of IoT Physical Devices.		
M 4.01	Basic Building blocks of an IoT Device	7	Applying
M 4.02	To learn about Exemplary Device	7	Understanding
	Series Test – II	1	
Contents: IoT Physical Devices and Endpoints What is an IoT device? , Basic Building blocks of an IoT Device, Exemplary Device: Raspberry Pi, About the Board, Linux on Raspberry Pi, Raspberry Pi Interfaces, Other IoT devices.			

Text / Reference

T/R	Book Title/Author
T1	Internet of Things – A Hands on Approach, By Arshdeep Bahga and Vijay Madisetti Universities Press, ISBN: 9788173719547

T2	Designing the Internet of Things – Adrian McEwen & Hakim Cassimally Wiley India, ISBN: 9788126556861
R1	The Internet of Things – Key Applications and Protocols, Wiley Publication, Olivier Hersent, David Boswarthick, Omar Elloumi. ISBN: 9788126557653
R2	The Internet of Things , Pearson, By Michael Miller ISBN: 9789332552456

Online Resources

Sl.No	Website Link
1	http://www.cisco.com/c/dam/en_us/solutions/trends/iot/introduction_to_IoT_november.pdf
2	https://www.ftc.gov/system/files/documents/reports/federal-trade-commission-staff-report-november-2013-workshop-entitled-internet-things-privacy/150127iotrpt.pdf